Reform and Practice of Teaching Mode for "Principles and Applications of Microcontrollers" in the Perspective of Digital and Intelligent Transformation

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Abstract

In the context of the rapid development of digitalization and intelligentization, the traditional teaching mode of the "Principles and Applications of Microcontrollers" course is facing challenges such as slow update of teaching content, unclear progression of abilities, difficulty in achieving personalized learning, weak process evaluation, and difficulty in evaluating the achievement of abilities. This paper proposes a new course teaching mode for the course, which is a four-fusion, three-progressive, and two-combination model. Based on the OBE concept as the overall design, it deeply integrates digitalization empowerment methods such as knowledge graphs, ability graphs, AI tutors, and digital virtual humans. In terms of teaching content, it further promotes the integration of production and education, science and education, competition and innovation, and innovation and entrepreneurship in the course. In terms of teaching methods, it innovates the deep blended teaching method based on ability progression of BOPPPS + ENGINE. In terms of teaching evaluation, it constructs a new evaluation system that combines process evaluation and Terminal evaluation, and dynamic tracking and precise feedback. The practical results show that the students in the experimental group have significantly improved their ability

to solve complex engineering problems, verifying the feasibility and effectiveness of the teaching model proposed in this paper in the digitalization context for course reform, and providing a practical path for improving the teaching quality of similar courses that can be widely promoted.